Genre-Conditioned Latent Diffusion

High-Fidelity Audio Generation via Lightweight Conditioning

Midterm Report

Deep Generative Models

The End-to-End Pipeline

Input

GTZAN Audio
(Augmented)

Encode

Mel-Spectrogram
→ VAE Latent

Diffuse

U-Net Denoising
+ Genre Embed

Decode

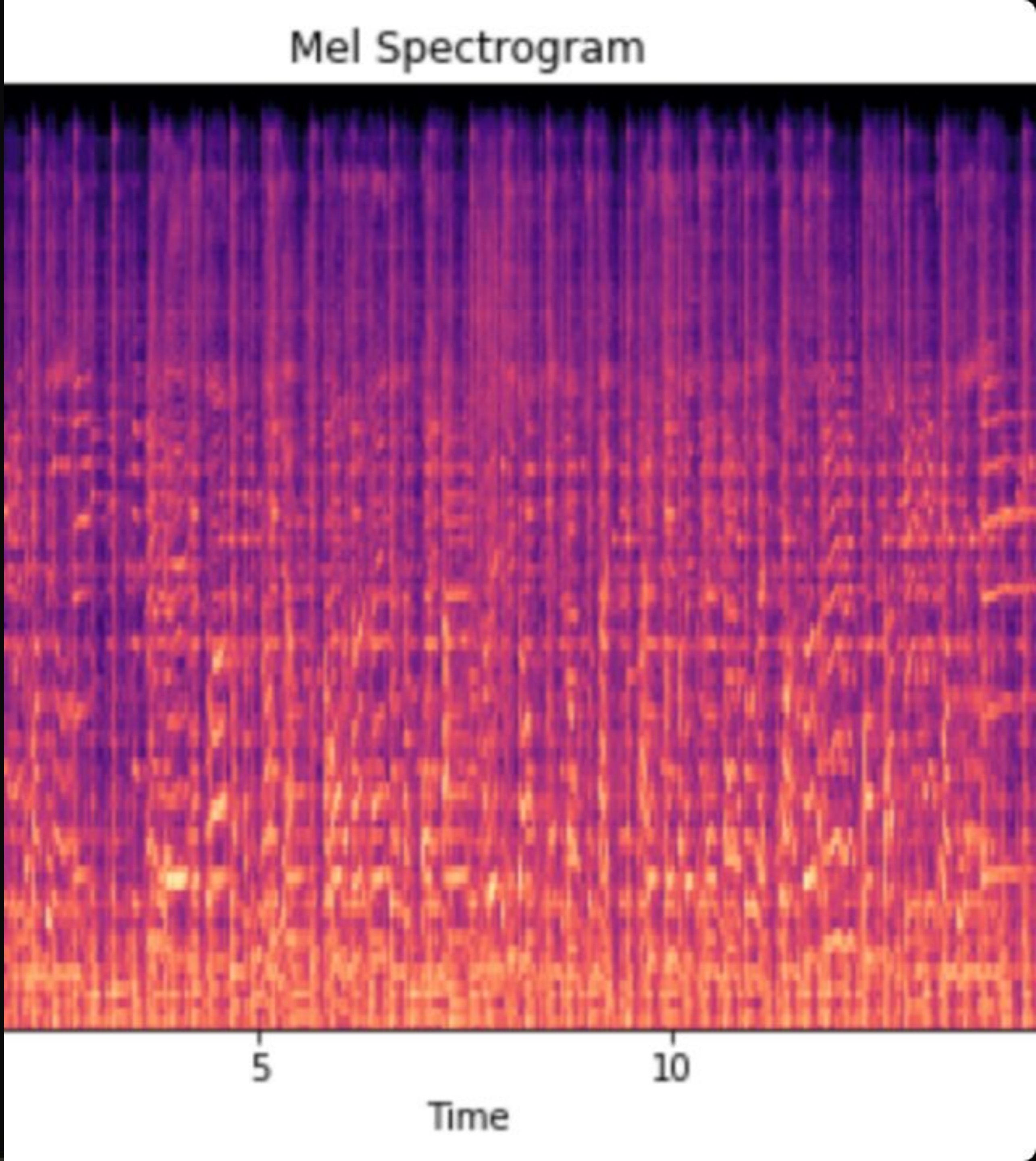
Latent → Mel
(VAE Decoder)

Vocode

Mel → Audio
(HiFi-GAN)

Data Strategy

- **Dataset:** GTZAN (1,000 tracks, 10 genres).
- **The Challenge:** Small dataset size risks severe overfitting.
- **The Solution:** Aggressive Data Augmentation.
  - Random Cropping (5s segments)
  - Time Stretching (±10%)
  - Pitch Shifting (±2 semitones)
- **Preprocessing:** Convert to Mel-Spectrograms using librosa before encoding.



Perceptual Compression (VAE)

The Encoder

We utilize a pre-trained **KL-Regularized VAE** (from AudioLDM/SD). This compresses the high-dimensional Mel-Spectrogram into a compact **Latent Space (z)**.

Benefit: Reduces computational complexity by ~64x, enabling faster training and inference.

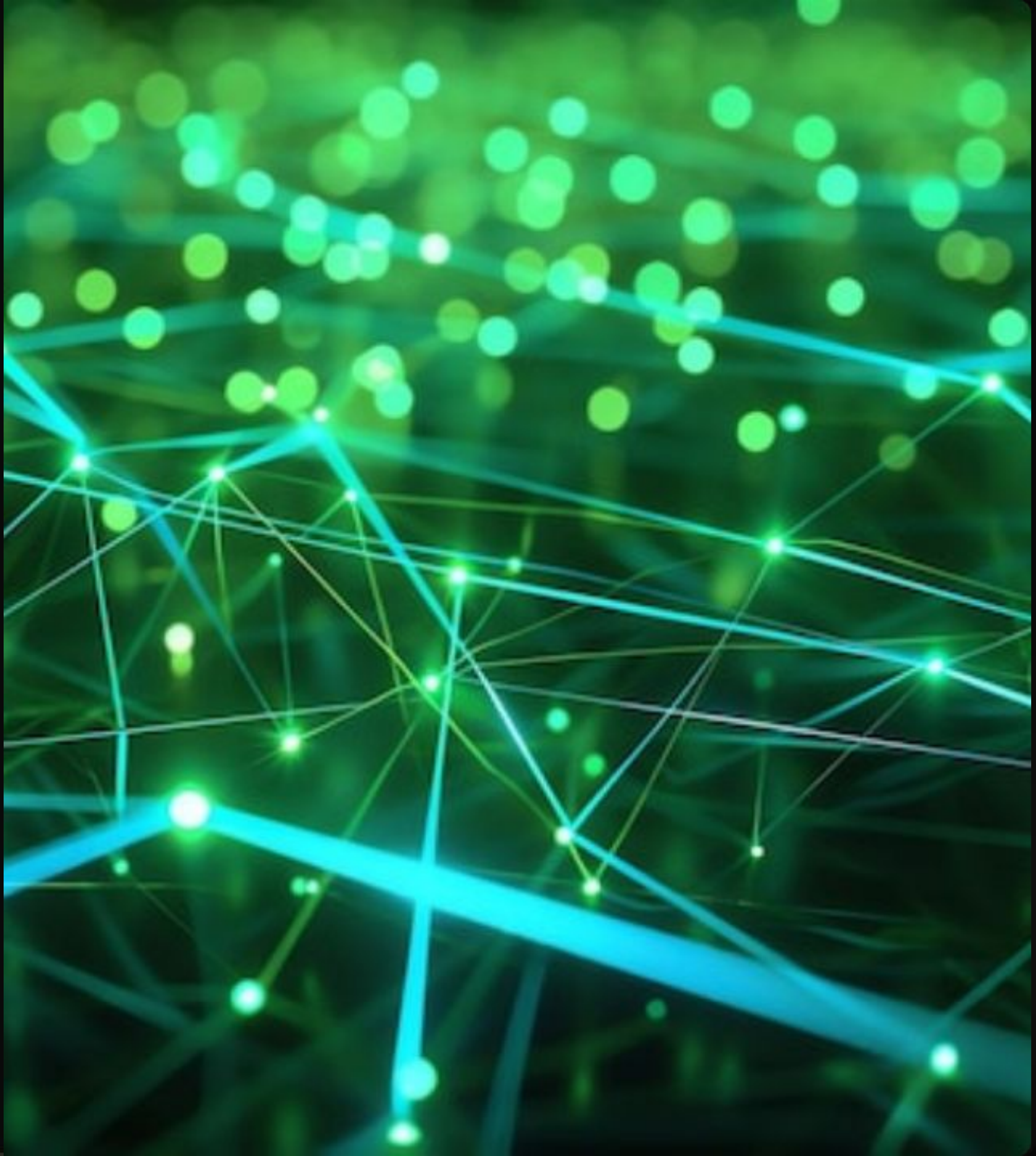
The Decoder

The decoder reconstructs the denoised latent vectors back into Mel-Spectrograms. This component is frozen during training.

Status: **Pre-trained Component** (No training required).

Novelty: Lightweight Conditioning

- **Standard Approach (AudioLDM):** Uses massive CLAP/T5 text encoders (100M+ params) to process prompts like "A jazz song".
- **Innovation:** A lightweight **Learnable Embedding Layer**.
  `nn.Embedding(10, 512)`
- **Efficiency:** Reduces parameter count drastically while maintaining precise control over the 10 GTZAN genres.
- **Mechanism:** Maps "Genre ID" directly to a rich



Core Architecture: The U-Net

Architecture Details

We are training a **Conditional U-Net** from scratch on the GTZAN latent embeddings.

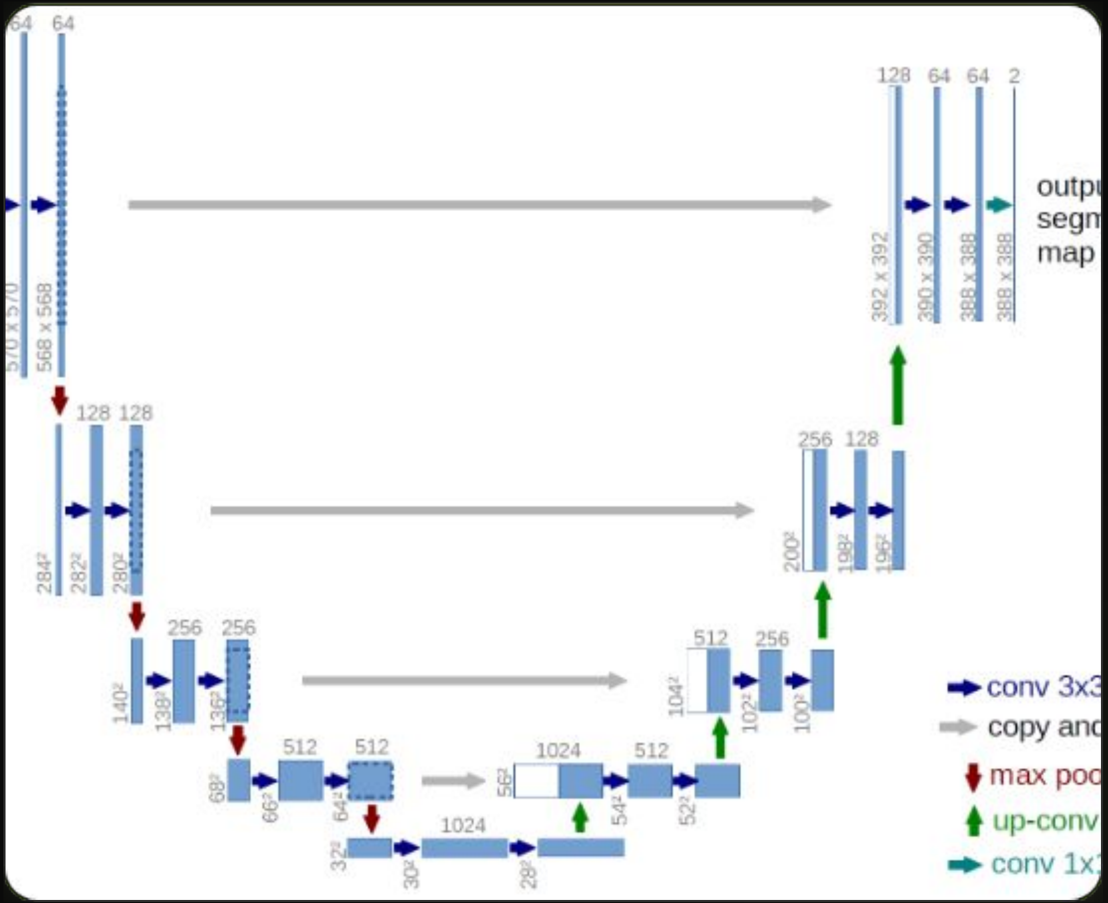
Cross-Attention Mechanism

This is how we inject the genre control:

Query (Q): The Noisy Audio Latent (z_t).

Key (K) & Value (V): The Genre Embedding Vector.

The model attends to the specific stylistic features of the genre at every spatial location of the spectrogram.



Training Strategy

→Forward Process

We gradually add Gaussian noise to the latent vectors over T steps until the data becomes pure isotropic noise.

←Reverse Process

The U-Net learns to predict the noise ϵ_θ added at each step, conditioned on the genre embedding.

Guidance

Classifier-Free Guidance: We randomly drop the genre label ($p=0.1$) during training. This amplifies genre adherence during inference.

Loss Function

Objective

We use the standard simple diffusion loss (MSE). The model tries to minimize the difference between the **actual noise** added and the **predicted noise**.

Optimization

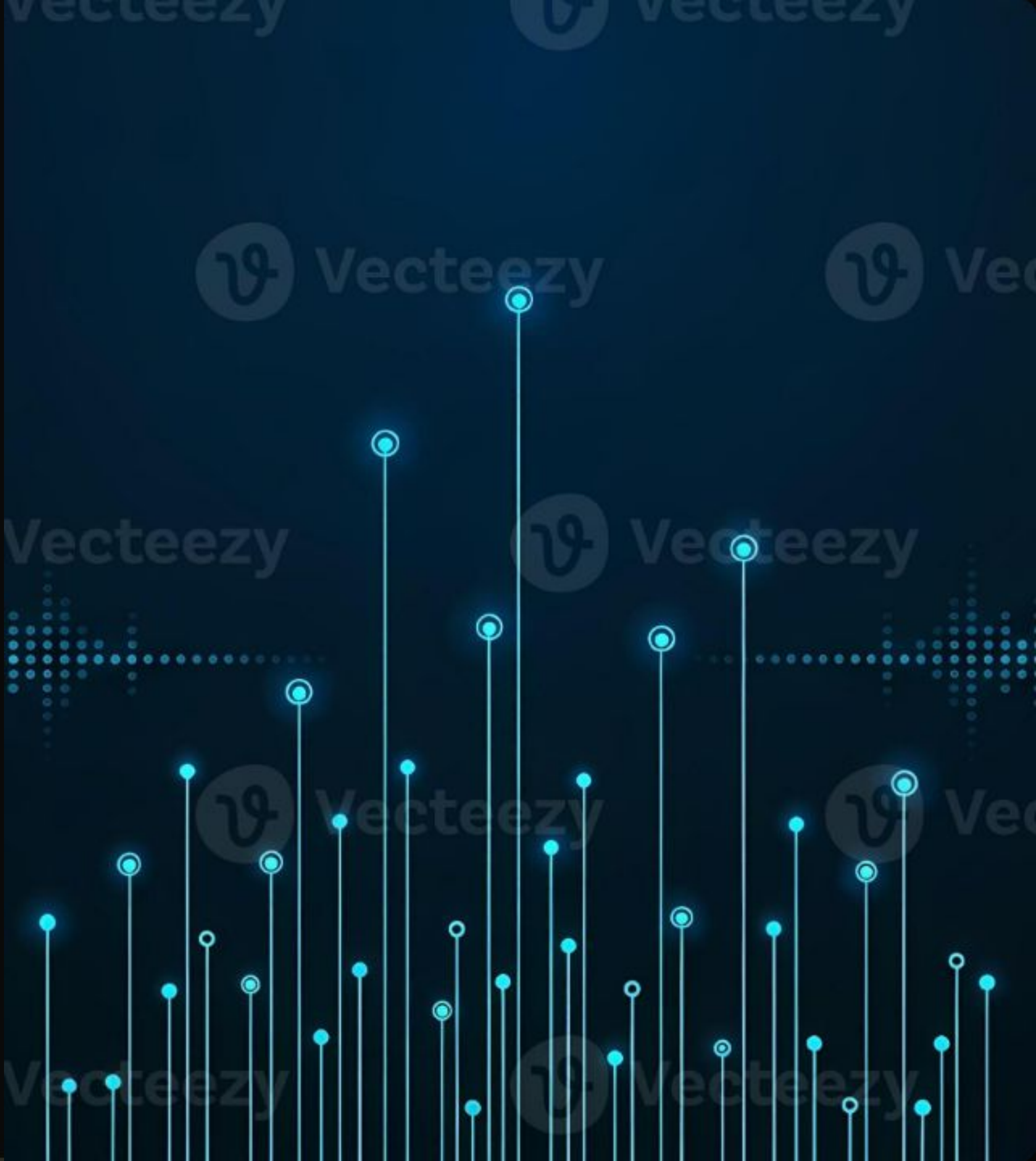
Optimizer: AdamW
Learning Rate: 1e-4
Batch Size: 16

$$L_{simple} = E_{z,c,t,\epsilon} \epsilon - \epsilon_{\theta} z_t t c^2$$

Where c is our Genre Embedding

The Vocoder: HiFi-GAN

- **Role:** Converts the reconstructed Mel-Spectrograms (images) back into raw Audio Waveforms (sound).
- **Architecture:** Generative Adversarial Network (GAN) optimized for high-fidelity audio synthesis.
- **Implementation:** We use a **Universal Pre-trained Checkpoint** (trained on LJSpeech/LibriTTS).
- **Why Pre-trained?** Training a vocoder from scratch is unstable and outside the scope of "Generation Control".



Comparative Advantages

⚡vs. Autoregressive (Jukebox)

Speed: Our Latent Diffusion model generates the entire spectrogram in parallel steps, avoiding the slow, sequential token prediction of Jukebox/MusicLM.

Compute: Runs on a single GPU vs. massive clusters.

🖼vs. Pixel Diffusion (Riffusion)

Semantics: Operating in Latent Space captures high-level semantic structure (melody/harmony) better than raw pixel space.

Quality: Avoids the "phase artifacts" and fuzzy sound common in Riffusion.

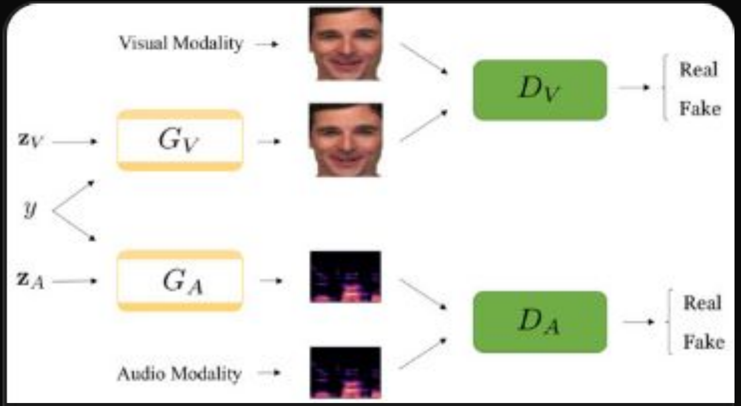
Real-World Applications

Royalty-Free Assets

Generating infinite, unique background tracks for content creators (YouTubers, Streamers) to avoid copyright strikes.

Adaptive Gaming Audio

Dynamic soundtracks that change genre based on gameplay state (e.g., Peaceful → Combat) in real-time.



Producer Tools

DAW plugins for "Idea Starting"—generating rhythmic or melodic foundations to break writer's block.

Project Roadmap

1

Data Prep

Augment GTZAN &
Cache Latents

2

Architecture

Build U-Net &
Embedding Layer

3

Training

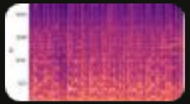
Run Training Loop
(50-100 Epochs)

4

Evaluation

Calc FAD Score &
Generate Samples

Image Sources



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